

## 1 Product Spaces

Given two measurable spaces  $(X, \mathcal{F})$  and  $(Y, \mathcal{G})$ , we can construct  $(X \times Y, \mathcal{F} \otimes \mathcal{G})$ , where the *measurable rectangles*  $\{A \times B \mid A \in \mathcal{F}, B \in \mathcal{G}\}$  generate the *product  $\sigma$ -algebra*

$$\mathcal{F} \otimes \mathcal{G} := \sigma(\{A \times B \mid A \in \mathcal{F}, B \in \mathcal{G}\}).$$

**Theorem 1.1.** Suppose  $(X, \sigma(\mathcal{A}))$  and  $(Y, \sigma(\mathcal{B}))$  are measurable spaces, where  $\emptyset \neq \mathcal{A} \subseteq \mathcal{P}(X)$  and  $\emptyset \neq \mathcal{B} \subseteq \mathcal{P}(Y)$ . Assume there exist  $\{E_n\}_{n \geq 1} \subseteq \mathcal{A}$  and  $\{F_n\}_{n \geq 1} \subseteq \mathcal{B}$  such that  $X = \bigcup_{n \geq 1} E_n$  and  $Y = \bigcup_{n \geq 1} F_n$ . Then

$$\sigma(\mathcal{A}) \otimes \sigma(\mathcal{B}) = \sigma(\{A \times B \mid A \in \mathcal{A}, B \in \mathcal{B}\}).$$

**Proof.** Define  $\mathcal{H} := \sigma(\{A \times B \mid A \in \mathcal{A}, B \in \mathcal{B}\})$ . Clearly,  $\mathcal{H} \subseteq \sigma(\mathcal{A}) \otimes \sigma(\mathcal{B})$ . It remains to prove the reverse inclusion.

*Step 1.* Fix  $B \in \mathcal{B}$  and define

$$\tilde{\mathcal{F}}_B := \{A \in \sigma(\mathcal{A}) \mid A \times B \in \mathcal{H}\}.$$

Since  $\mathcal{A} \subseteq \tilde{\mathcal{F}}_B$ , it is enough to show that  $\tilde{\mathcal{F}}_B$  is a  $\sigma$ -algebra. For  $A_1, A_2, \dots \in \tilde{\mathcal{F}}_B$ ,

$$\begin{aligned} \left( \bigcup_{n=1}^{\infty} A_n \right) \times B &= \bigcup_{n=1}^{\infty} (A_n \times B) \in \mathcal{H} \\ \implies \bigcup_{n=1}^{\infty} A_n &\in \tilde{\mathcal{F}}_B \end{aligned}$$

Also, since  $X = \bigcup_{n=1}^{\infty} E_n$ ,

$$X \times B = \bigcup_{n=1}^{\infty} (E_n \times B) \in \mathcal{H}.$$

Thus, for  $A \in \tilde{\mathcal{F}}_B$ ,

$$A^c \times B = (X \times B) \setminus (A \times B) \in \mathcal{H},$$

so  $A^c \in \tilde{\mathcal{F}}_B$ . Therefore  $\tilde{\mathcal{F}}_B$  is a  $\sigma$ -algebra and  $\sigma(\mathcal{A}) = \tilde{\mathcal{F}}_B$ . Hence

$$A \times B \in \mathcal{H} \quad \forall A \in \sigma(\mathcal{A}), B \in \mathcal{B}.$$

*Step 2.* Fix  $A \in \sigma(\mathcal{A})$  and define

$$\tilde{\mathcal{G}}_A := \{B \in \sigma(\mathcal{B}) \mid A \times B \in \mathcal{H}\}.$$

By Step 1,  $\mathcal{B} \subseteq \tilde{\mathcal{G}}_A$ . We show that  $\tilde{\mathcal{G}}_A$  is a  $\sigma$ -algebra. If  $B_1, B_2, \dots \in \tilde{\mathcal{G}}_A$ , then

$$\begin{aligned} A \times \left( \bigcup_{n=1}^{\infty} B_n \right) &= \bigcup_{n=1}^{\infty} (A \times B_n) \in \mathcal{H} \\ \implies \bigcup_{n=1}^{\infty} B_n &\in \tilde{\mathcal{G}}_A \end{aligned}$$

Also, since  $Y = \bigcup_{n=1}^{\infty} F_n$  and  $F_n \in \mathcal{B}$ , Step 1 gives

$$A \times Y = \bigcup_{n=1}^{\infty} (A \times F_n) \in \mathcal{H}.$$

Thus, for  $B \in \tilde{\mathcal{G}}_A$ ,

$$A \times B^c = (A \times Y) \setminus (A \times B) \in \mathcal{H},$$

so  $B^c \in \tilde{\mathcal{G}}_A$ . Therefore is a  $\sigma$ -algebra. Since,  $\sigma(\mathcal{B}) = \tilde{\mathcal{G}}_A$ . Hence

$$A \times B \in \mathcal{H} \quad \forall A \in \sigma(\mathcal{A}), B \in \sigma(\mathcal{B}).$$

Therefore  $\sigma(\mathcal{A}) \otimes \sigma(\mathcal{B}) \subseteq \mathcal{H}$ . Combining both inclusions,

$$\sigma(\mathcal{A}) \otimes \sigma(\mathcal{B}) = \sigma(\{A \times B \mid A \in \mathcal{A}, B \in \mathcal{B}\}).$$

□

**Remark 1.2.** The argument is a two-stage extension. First enlarge the  $A$ -coordinate from  $\mathcal{A}$  to  $\sigma(\mathcal{A})$  while keeping  $B \in \mathcal{B}$  fixed. Then enlarge the  $B$ -coordinate from  $\mathcal{B}$  to  $\sigma(\mathcal{B})$ . The countable covers of  $X$  and  $Y$  are exactly what allow the complement property to be proved.

## 2 Finite Products

**Definition 2.1.** Let  $(X_1, \mathcal{F}_1), \dots, (X_n, \mathcal{F}_n)$  be  $n$  measurable spaces. Their product is

$$(X_1, \mathcal{F}_1) \times \dots \times (X_n, \mathcal{F}_n) = \left( \prod_{i=1}^n X_i, \mathcal{F}_1 \otimes \dots \otimes \mathcal{F}_n \right),$$

where the product  $\sigma$ -algebra is

$$\mathcal{F}_1 \otimes \dots \otimes \mathcal{F}_n = \sigma(\{A_1 \times \dots \times A_n \mid A_i \in \mathcal{F}_i, 1 \leq i \leq n\}).$$

**Corollary 2.2.** Inductively, we can define

$$\underbrace{\mathcal{B}_{\mathbb{R}} \otimes \dots \otimes \mathcal{B}_{\mathbb{R}}}_{d \text{ times}} = \mathcal{B}_{\mathbb{R}^d}.$$

## 3 Sections of Product Sets

Let  $E \subseteq X \times Y$ . For  $x \in X$ , define  $E^x := \{y \in Y \mid (x, y) \in E\}$ . This is called the *vertical section* of  $E$  at the horizontal coordinate  $x$ .

Similarly, for  $y \in Y$ , define  $E_y := \{x \in X \mid (x, y) \in E\}$ . This is called the *horizontal section* of  $E$  at the vertical coordinate  $y$ .

**Remark 3.1.** For a measurable rectangle  $A \times B$ ,

$$(A \times B)^x = \begin{cases} B, & x \in A, \\ \emptyset, & x \notin A, \end{cases}$$

and

$$(A \times B)_y = \begin{cases} A, & y \in B, \\ \emptyset, & y \notin B. \end{cases}$$

**Theorem 3.2.** Let  $(X, \mathcal{F})$  and  $(Y, \mathcal{G})$  be measurable spaces, and let  $E \in \mathcal{F} \otimes \mathcal{G}$ . Then

$$E_x \in \mathcal{G} \quad \forall x \in X \tag{1}$$

$$E_y \in \mathcal{F} \quad \forall y \in Y \tag{2}$$

**Proof.** For fixed  $x \in X$ , define  $T^x : Y \rightarrow X \times Y$  with  $T^x(y) = (x, y)$ . For  $A \in \mathcal{F}$  and  $B \in \mathcal{G}$ ,

$$(T^x)^{-1}(A \times B) = \begin{cases} B, & x \in A, \\ \emptyset, & x \notin A. \end{cases} \in \mathcal{G}.$$

Since measurable rectangles generate  $\mathcal{F} \otimes \mathcal{G}$ ,  $T^x$  is measurable. Hence  $E^x = (T^x)^{-1}(E) \in \mathcal{G}$ . Similarly, for fixed  $y \in Y$ , define  $T_y : X \rightarrow X \times Y$  with  $T_y(x) = (x, y)$ . Then

$$(T_y)^{-1}(A \times B) = \begin{cases} A, & y \in B, \\ \emptyset, & y \notin B, \end{cases} \in \mathcal{F},$$

so  $T_y$  is measurable. Therefore  $E_y = (T_y)^{-1}(E) \in \mathcal{F}$  □

**Remark 3.3.** The converse need not hold. In general,

$$[E^x \in \mathcal{G} \ \forall x \in X \quad \text{and} \quad E_y \in \mathcal{F} \ \forall y \in Y] \not\Rightarrow E \in \mathcal{F} \otimes \mathcal{G}.$$

Indeed,

$$E = \bigcup_{x \in X} (\{x\} \times E^x) = \bigcup_{y \in Y} (E_y \times \{y\}),$$

and these are in general uncountable unions, whereas a  $\sigma$ -algebra is only required to be closed under countable unions.

**Theorem 3.4.** Let  $(\Omega, \mathcal{M})$  be a measurable space and let  $f : X \times Y \rightarrow \Omega$  be  $\mathcal{F} \otimes \mathcal{G} / \mathcal{M}$  measurable. Then, for every  $x \in X$ , the function  $f^x : Y \rightarrow \Omega$  with  $f^x(y) := f(x, y)$  is  $\mathcal{G} / \mathcal{M}$  measurable.

Similarly, for every  $y \in Y$ ,  $f_y : X \rightarrow \Omega$  with  $f_y(x) := f(x, y)$  is  $\mathcal{F} / \mathcal{M}$  measurable.

**Proof.** For every  $x \in X$ ,  $f^x = f \circ T^x$  and for every  $y \in Y$ ,  $f_y = f \circ T_y$ . Since  $T^x$  and  $T_y$  are measurable, both  $f^x$  and  $f_y$  are measurable. □